

Advanced (Habanero)

- Q1.** What is the vertex form of a quadratic function?
(A) $y = a(x - h)^2 + k$
(B) $y = a(x + h)^2 - k$
(C) $y = ax^2 + bx + c$
(D) $y = a(x - h)(x - k)$
(E) $y = ax^2 - bx - c$
- Q2.** Which of the following is the intercept form of $y = x^2 - 4x - 5$?
(A) $y = (x - 5)(x + 1)$
(B) $y = (x + 5)(x - 1)$
(C) $y = (x - 2)(x - 3)$
(D) $y = (x + 2)(x + 3)$
(E) $y = (x - 1)(x + 5)$
- Q3.** What is the vertex of the parabola $y = x^2 - 6x + 8$?
(A) (0, 8)
(B) (3, -1)
(C) (2, 4)
(D) (3, 1)
(E) (0, 0)
- Q4.** If $y = a(x - h)^2 + k$, what does (h, k) represent?
(A) The y -intercept and x -intercept
(B) The vertex of the parabola
(C) The x -intercepts of the parabola
(D) The y -intercept and the axis of symmetry
(E) The axis of symmetry and the y -intercept
- Q5.** What is the equation of the axis of symmetry for the parabola $y = x^2 - 4x - 3$?
(A) $x = -2$
(B) $x = 2$
(C) $x = 4$
(D) $x = 1$
(E) $x = 3$
- Q6.** If $y = (x - 3)(x + 1)$, what are the x -intercepts of the parabola?
(A) (-3, 0) and (1, 0)
(B) (-1, 0) and (3, 0)
(C) (-1, 0) and (-3, 0)
(D) (1, 0) and (3, 0)
(E) (-2, 0) and (4, 0)
- Q7.** What is the y -intercept of $y = x^2 - 6x + 8$?
(A) (-6, 0)
(B) (0, 8)
(C) (0, -8)
(D) (8, 0)
(E) (6, 0)
- Q8.** If $y = (x + 2)(x - 5)$, what is the equation in vertex form?
(A) $y = (x - 1.5)^2 - 8.25$
(B) $y = (x + 2)(x - 5)$
(C) $y = x^2 - 3x - 10$
(D) $y = x^2 - 6x + 8$
(E) $y = x^2 + 3x + 2$

Open Ended Questions (FRQ)

- Q9.** Write $y = x^2 - 8x + 12$ in vertex form and find the vertex of the parabola
- Q10.** Write $y = (x + 3)(x - 2)$ in vertex form and find the y -intercept of the parabola

Chef's Tips

- Tip 1: Please ensure students have a graphing calculator
- Tip 2: Remind students to check their work carefully
- Tip 3: Encourage students to use the given formulas to simplify their work